

The Human Hand as a Tool

People use their hands for many things. People can type on a computer keyboard, wave at a friend, use a cellphone, play video games, and point to give directions. A human hand is especially useful because the fingers and thumb work together to make complex movements. This structure allows us to grip and hold objects, build things, use a pen to write, and play sports and musical instruments.

There are several elements that enable our hands to do so many things. The hand has joints, places where two bones come together. The **joints** in our hands allow fingers to bend, which lets our hands open and close. The **tendons** that run along each finger work like a string to pull the fingers in toward the palm. They are connected to **muscle**, which powers the tendons, allowing the hand to grip with force when it closes. Our hands are covered with **skin** that has enough friction so that items we grip do not slip away. And the wrist can **swivel**, or turn back and forth. Because of these interconnected parts in the hand, our hands can squeeze, tap, bend, lift, push, pull, and press.

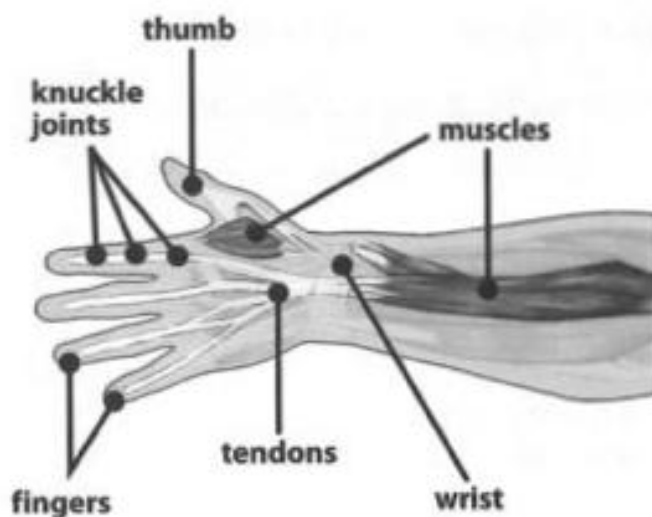
A tool designed to work like a hand might use a lever. A **lever** is a simple machine that consists of something rigid, such as a bar, a board, or a bone, and a fulcrum. The **fulcrum** can be a hinge, an axle, an elbow, or just something to push against to move. A lever is used to help lift or move a load when force is applied. Levers are often used in pairs to work against each other, as in scissors.

A lever gives the user **mechanical advantage**, or additional force for the same amount of effort. How much force depends on where the fulcrum and load are and where the force is applied. Each of these elements will be at one end or the other and somewhere in the middle.

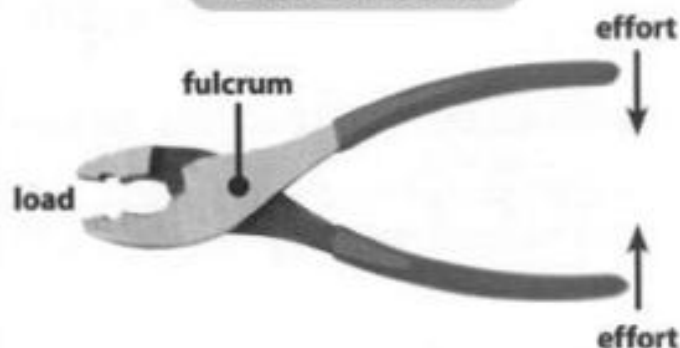
STEM Challenge: Helping Hand

Visual Literacy

Parts of a human hand



Parts of a lever



fulcrum: the place where the lever or levers are free to rotate

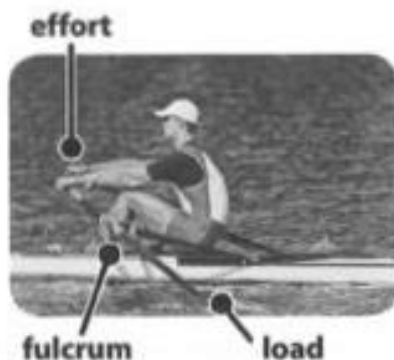
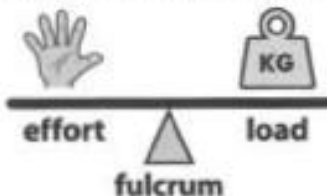
load: the place that holds, carries, or moves the weight

effort: the place where you apply force

Classes of levers

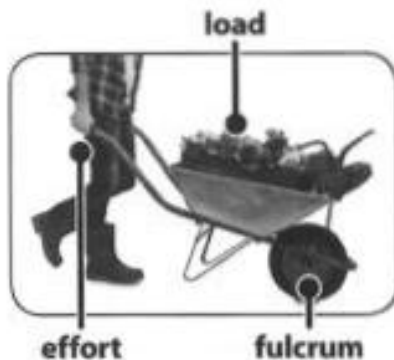
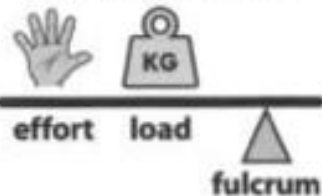
1st class lever

fulcrum in the middle



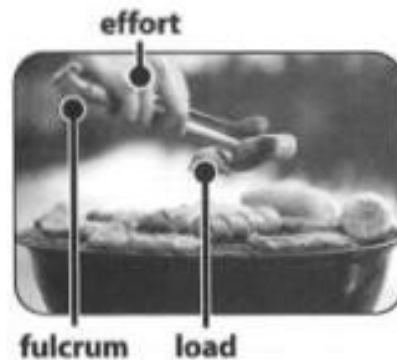
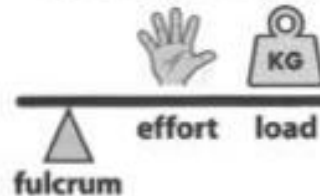
2nd class lever

load in the middle



3rd class lever

effort in the middle



Helping Hand

Challenge: Design a helping hand that can pick up, hold, and release items.

Testable goals: The hand tool picks up 5 different items, holds each for 5 seconds, and then releases them safely.

Research: Look at pictures of levers, their parts, and other simple machines. Notice how the devices are constructed. Think about how science, technology, engineering, and math are used to create these machines.

Brainstorm: Draw one or more design ideas for a helping hand. There are many different ways to complete this challenge. Be creative!

A large, empty rectangular box with a double-lined border, intended for students to draw their design ideas for a helping hand.

STEM Challenge: Helping Hand

Suggested Materials List

Item for each group

- ☐ scissors

Items for the whole class

- ☐ craft sticks
- ☐ straws
- ☐ wooden dowels
- ☐ metal brads
- ☐ rulers
- ☐ sponge pieces
- ☐ small balloons
- ☐ masking tape
- ☐ binder clips
- ☐ rubber bands
- ☐ glue
- ☐ string

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Items for testing

- ☐ 8 items of various sizes, shapes, and textures to pick up

1 Plan

Think about the materials available to you. How can you use the materials to create a helping hand that is able to pick up, hold, and release objects?



2 Create

Use the materials you have to create a helping hand. Follow your design as closely as possible.



3 Test

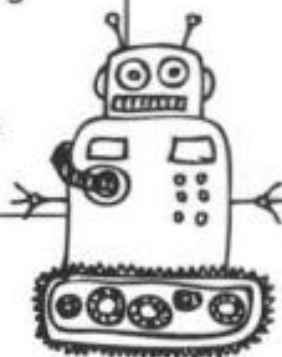
Use your helping hand. Can it pick up different-sized objects? Can it hold them? Can it let go of the objects?



4 Evaluate and Revise

Evaluate the performance of your helping hand. What revisions can you make to improve its performance?

Repeat the design process until you are satisfied with your helping hand.



STEM Challenge: Helping Hand

Design Process

1 Plan: Describe and draw your design. List your materials.



2 Create



3 Test: Describe and draw your results.



4 Evaluate

1 Redesign: Describe what you will change. Draw your new design.



2 Revise or re-create



3 Test: Describe and draw your results.



4 Evaluate